Lecture 2

One state, many preparations

Objectives. *A density operator fixes expectations, not a preparation history. The extreme points of the set of density operators are exactly the rays. When* $\dim \mathcal{H} \geq 2$*, even a ray leaves some bounded observable uncertain. A unitary on padded preparation labels relates any two finite pure-state ensembles of the same state. For one optical mode, averaging the phase of a coherent state gives exactly a Poisson mixture of number states.*

- Prove that the extreme density operators have rank one, and find an uncertain event in any state when $\dim \mathcal{H} \geq 2$.
- Construct the unitary relating two finite ensembles of the same state.
- Derive the phase-averaged coherent-state identity and explain what it does and does not say about a laser.

2.1 What mixing forgets

Theorem 1.7 identified a state with the unique density operator that gives its bounded expectations. Uniqueness of the operator does not mean uniqueness of the preparation. In *QM I*, §3.6, half the spins prepared up and half down along z gave the same state as a preparation using the two x directions. The same distinction survives in infinite dimension. We keep the nonzero separable complex Hilbert space $\mathcal{H}$ of Lecture 1 and write $\mathcal{D}(\mathcal{H})$ for its density operators.

Definition 2.1 (Convexity and pure states)**.** A set is *convex* if it contains $t\rho_1 + (1-t)\rho_2$ whenever it contains ρ_1, ρ_2 and $0 \leq t \leq 1$. A point ρ in a convex set is *extreme* if, for ρ_1, ρ_2 belonging to that same set,

$$\rho = t\rho_1 + (1-t)\rho_2, \quad 0 < t < 1 \quad \Longrightarrow \quad \rho_1 = \rho_2 = \rho.$$

A *pure state* is an extreme point of $\mathcal{D}(\mathcal{H})$; a state that is not pure is *mixed*.

The density operators form a convex set: a convex combination is bounded and positive, and its diagonal sum in any ONB is $t \operatorname{Tr} \rho_1 + (1-t) \operatorname{Tr} \rho_2 = 1$. Its mean of a bounded A is $t \operatorname{Tr}(\rho_1 A) + (1-t) \operatorname{Tr}(\rho_2 A)$, by the basis expression for the trace proved in § 1.2. Physically, select one preparation with probability t and the other with probability $1-t$, and discard the selection record. Retaining that record is additional information about the preparation, not part of the density operator of this system alone.

The restriction $0 < t < 1$ matters. A pure state can occur in a mixture of two identical copies of itself; extremality only excludes a nontrivial mixture of *different* states. Nor does an extreme point have to be isolated: the pure qubit states form the boundary sphere of the qubit's Bloch ball. The question is whether a line segment joining different states can have this point strictly between its endpoints.

Self-check 2.1

1. Is $\rho = \frac{1}{2}\rho + \frac{1}{2}\rho$ evidence that ρ is mixed?
2. Which step in the convexity argument would fail for weights $t = 2$ and $1 - t = -1$?

2.2 Pure means a ray

Theorem 2.2 (The extreme density operators). *For a density operator on $\mathcal{H}$, the following are equivalent: it is pure; it has rank one; it equals $|\psi\rangle\langle\psi|$ for a unit vector ψ. Two unit vectors give the same rank-one state exactly when they differ by a complex phase.*

Proof. A state of rank greater than one is not extreme. The spectral decomposition from Proposition 1.4 is $\rho = \sum_j p_j |e_j\rangle\langle e_j|$, with $p_j \geq 0$ and $\sum_j p_j = 1$. If more than one eigenvalue is positive, choose p_1 with $0 < p_1 < 1$ and set

$$\sigma = \frac{\rho - p_1 |e_1\rangle\langle e_1|}{1 - p_1} = \sum_{j \neq 1} \frac{p_j}{1 - p_1} |e_j\rangle\langle e_j| .$$

The last expression defines a positive bounded operator of trace one. Indeed, the positive partial sums converge in operator norm because their norm tails are bounded by tails of the summable weights; positivity passes to the limit. Its eigenbasis diagonal sum is $\sum_{j \neq 1} p_j/(1 - p_1) = 1$. Thus $\rho = p_1 |e_1\rangle\langle e_1| + (1 - p_1)\sigma$ is a convex decomposition into different states: $\sigma e_1 = 0$, whereas $|e_1\rangle\langle e_1| e_1 = e_1$. Consequently an extreme state has just one nonzero eigenvalue, which must be 1. It is $|\psi\rangle\langle\psi|$ for a unit eigenvector ψ.

A rank-one state is extreme. Suppose $|\psi\rangle\langle\psi| = t\rho_1 + (1 - t)\rho_2$ with $0 < t < 1$. For any $v \perp \psi$,

$$0 = t \langle v, \rho_1 v\rangle + (1 - t) \langle v, \rho_2 v\rangle .$$

Both terms are nonnegative, so each is zero. The positive square root (S6) gives $\left\|\sqrt{\rho_i} v\right\|^2 = 0$, hence $\rho_i v = 0$ for $i = 1, 2$. Self-adjointness then gives $\langle v, \rho_i \psi\rangle = \langle \rho_i v, \psi\rangle = 0$ for every $v \perp \psi$. Thus $\rho_i \psi = c_i \psi$ and $\rho_i = c_i |\psi\rangle\langle\psi|$; its trace is one, so $c_i = 1$. This proves extremality. A positive rank-one trace-one operator is necessarily such a projection, by its single nonzero eigenvalue.

Finally, if $|\psi\rangle\langle\psi| = |\varphi\rangle\langle\varphi|$, applying both sides to φ shows that $\varphi = \langle\psi, \varphi\rangle\, \psi$. The coefficient has modulus one because both vectors are unit vectors. The converse follows from $|c\psi\rangle\langle c\psi| = |c|^2 |\psi\rangle\langle\psi|$. ∎

This proof used the compact spectral theorem already supplied by L1, not a finite-dimensional restriction. An oscillator wave packet defines a pure state just as a spinor does. Its ray is now recovered from the convex structure of all normal states, rather than postulated as the only kind of state.

The familiar purity test also survives. The square ρ^2 is positive trace class: it has eigenvalues p_j^2, and $0 \leq p_j^2 \leq p_j$ makes their sum finite. Thus $\mathrm{Tr}(\rho^2) = \sum_j p_j^2 \leq \sum_j p_j = 1$. Equality holds precisely when $\sum_j p_j(1 - p_j) = 0$. Every term is nonnegative, so every p_j is either 0 or 1; exactly one is 1. Thus $\mathrm{Tr}(\rho^2) = 1$ if and only if the state is pure. In the qubit Bloch ball this is $\mathrm{Tr}(\rho^2) = (1 + |\mathbf{r}|^2)/2$, the boundary-sphere criterion from QM-I.

Self-check 2.2

1. In the rank-one proof, why does $\langle v, \rho_i v\rangle = 0$ imply $\rho_i v = 0$?
2. Can an infinite-rank density operator be pure? Identify the step that decides this.

2.3 Pure does not mean sharp for everything

For bounded self-adjoint A, its variance in state ρ is

$$\mathrm{Var}_\rho(A) = \mathrm{Tr}(\rho A^2) - [\mathrm{Tr}(\rho A)]^2 = \mathrm{Tr}\big(\rho(A - \mathrm{Tr}(\rho A)\mathbb{1})^2\big) \geq 0.$$

It is the mean square deviation of its readings. Here A is bounded, so both expectations fall within L1's theorem and exist without any extra moment assumption. A state is *dispersion-free for A* when this variance vanishes, and *dispersion-free* when it vanishes for every bounded self-adjoint A.

Proposition 2.3 (No universally sharp quantum state)**.** *If $\dim \mathcal{H} \geq 2$, every density operator gives some rank-one projection E a strictly positive variance. No state is dispersion-free on all bounded observables.*

Proof. Since $E^2 = E$, put $w = \mathrm{Tr}(\rho E)$; then $\mathrm{Var}_\rho(E) = w(1-w)$. If ρ is mixed, Theorem 2.2 gives at least two positive eigenvalues, each less than one. Choose one, $p \in (0,1)$, and a unit eigenvector e. For $E = |e\rangle\langle e|$, $w = p$ lies strictly between 0 and 1.

If $\rho = |\psi\rangle\langle\psi|$ is pure, choose a unit $\varphi \perp \psi$, possible because $\dim \mathcal{H} \geq 2$. Let $\chi = (\psi+\varphi)/\sqrt{2}$ and $E = |\chi\rangle\langle\chi|$. Its probability is $w = |\langle \chi, \psi\rangle|^2 = 1/2$, so its variance is $1/4$. ∎

The order of quantifiers is important: *for every state there is an uncertain event*. It does not say that one fixed projection is uncertain in every state. A unit vector in its range gives it probability one; a unit vector in its kernel gives probability zero. On a one-dimensional space every bounded observable is a scalar and the unique state is dispersion-free, which explains the dimension hypothesis.

For example, a spin prepared in $|0\rangle$ has zero variance for σ_z and variance one for σ_x. Being pure excludes uncertainty arising from a nontrivial convex preparation of that state; it does not exclude uncertainty in a particular reading. This result concerns quantum states and their expectations. Whether an enlarged theory can attach further values to a system is a different question, taken up in Lecture 13. No claim about all such theories has been proved here.

Self-check 2.3

1. In a pure state $|\psi\rangle\langle\psi|$, find an event whose probability is $1/2$.
2. Does the proposition say that every observable has nonzero variance in every state?

2.4 The unitary freedom of finite ensembles

Definition 2.4 (An ensemble of a state)**.** An *ensemble of ρ* is a finite or countable list $\{(p_i, \psi_i)\}$ of probabilities and unit vectors such that $\sum_i p_i = 1$ and $\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|$. The vectors need not be orthogonal or distinct. Zero-probability entries may be omitted.

For a countable list this series converges in operator norm, since the norm of its tail is at most $\sum_{i>N} p_i$. Its limit is positive with trace one: summing its diagonal entries interchanges only nonnegative terms, as in L1. Every state has at least its spectral ensemble. A finite ensemble, however, has finite-rank density operator even when the ambient Hilbert space is infinite-dimensional. The next theorem concerns *finite ensembles only*.

Absorb each weight into its vector, writing $x_i = \sqrt{p_i}\psi_i$. Then $\rho = \sum_i |x_i\rangle\langle x_i|$, with $\sum_i \|x_i\|^2 = 1$. These vectors are not generally unit vectors. A zero-weight entry is the zero vector; recovering a unit vector from it is unnecessary.

Theorem 2.5 (Hughston–Jozsa–Wootters, finite-ensemble form)**.** *Let $(x_i)_{i=1}^m$ and $(y_j)_{j=1}^m$ be two finite lists in $\mathcal{H}$ with $\sum_i \|x_i\|^2 = \sum_j \|y_j\|^2 = 1$, padding the shorter original list with zero vectors to get equal lengths. They generate the same density operator if and only if there is an $m \times m$ unitary matrix u with*

$$x_i = \sum_{j=1}^{m} u_{ij} y_j \qquad (1 \le i \le m). \tag{2.1}$$

Proof. A unitary relation gives the same operator. Expand the rank-one sums:

$$\sum_i |x_i\rangle\langle x_i| = \sum_{j,k} \Big(\sum_i u_{ij} \overline{u_{ik}} \Big) |y_j\rangle\langle y_k| = \sum_j |y_j\rangle\langle y_j| .$$

The coefficient is δ_{jk} because the columns of u are orthonormal.

Equal operators give a unitary relation. Suppose both sums equal ρ. Its range is the span of either list. Indeed, for $\xi \in \mathcal{H}$, $\langle \xi, \rho\xi \rangle = \sum_i |\langle x_i, \xi \rangle|^2$ shows that $\ker \rho = \operatorname{span}\{x_i\}^\perp$. This finite span is closed. The formula $\rho\xi = \sum_i x_i \langle x_i, \xi \rangle$ shows that ρ maps the span into itself. Its restriction has zero kernel; an injective map of a finite-dimensional space to itself is surjective, so it is invertible there. This proves the range claim, also for the y list. Write this common range as K and let $e_1, \dots, e_r$ be an eigenbasis on it, with $\rho e_a = \lambda_a e_a$, $\lambda_a > 0$. Necessarily $r \le m$.

For each a make two vectors in the *label space* $\mathbb{C}^m$. Here a labels a physical direction in K, whereas i labels a preparation entry. The vector v_a collects the coefficients of that one physical direction across all entries. Dividing by $\sqrt{\lambda_a}$ will make these label vectors orthonormal:

$$v_a = \left(\frac{\langle e_a, x_i \rangle}{\sqrt{\lambda_a}} \right)_{i=1}^{m}, \qquad w_a = \left(\frac{\langle e_a, y_j \rangle}{\sqrt{\lambda_a}} \right)_{j=1}^{m}.$$

The eigenbasis matrix entries of ρ give

$$\sum_i \langle e_a, x_i \rangle \, \overline{\langle e_b, x_i \rangle} = \langle e_a, \rho e_b \rangle = \lambda_b \delta_{ab}.$$

Taking complex conjugates and dividing by $\sqrt{\lambda_a \lambda_b}$ shows $\langle v_a, v_b \rangle = \delta_{ab}$. The same calculation gives $\langle w_a, w_b \rangle = \delta_{ab}$. Thus each is an orthonormal r-frame in the same $\mathbb{C}^m$. Complete them to ONBs $(v_a)_{a=1}^m$, $(w_a)_{a=1}^m$ and set $u = \sum_{a=1}^m |v_a\rangle\langle w_a|$, the unitary sending each w_a to v_a. In coordinate i, $(v_a)_i = (u w_a)_i = \sum_j u_{ij} (w_a)_j$. Multiplying by $\sqrt{\lambda_a}$ gives, for $1 \le a \le r$,

$$\langle e_a, x_i \rangle = \sum_j u_{ij} \langle e_a, y_j \rangle .$$

All the x_i, y_j lie in K, so agreement of these r coordinates gives (2.1). Only finite-dimensional ONB completion was used; there is no inverse of ρ on the whole Hilbert space in this argument. ∎

The unitary acts on the list of preparation labels. It is not a unitary operation on the physical system sending the unit vectors of one ensemble to those of the other. Such an operation would preserve inner products; the new ensemble need not be orthogonal even if the original one was. The theorem classifies finite descriptions of a common state, not a transformation between distinct states.

The two decompositions in the following example appear as different segments through one Bloch point in Figure 2.1.

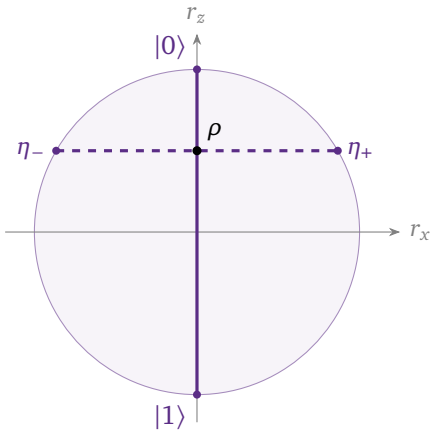


Figure 2.1: The section $r_y = 0$ of the qubit Bloch ball. The spectral endpoints with weights $3/4, 1/4$ and the nonorthogonal endpoints η_+, η_- with weights $1/2, 1/2$ give the same interior point $\mathbf{r} = (0, 0, 1/2)$; the segments describe mixtures, not physical trajectories.

For a concrete example (Nielsen–Chuang, p. 103), consider

$$\rho = \frac{3}{4}\,|0\rangle\langle 0| + \frac{1}{4}\,|1\rangle\langle 1| = \frac{1}{2}\,|\eta_+\rangle\langle\eta_+| + \frac{1}{2}\,|\eta_-\rangle\langle\eta_-|\,, \qquad \eta_\pm = \frac{\sqrt{3}\,|0\rangle \pm |1\rangle}{2}. \tag{2.2}$$

The vectors $\eta_\pm$ are unit vectors with overlap $1/2$, so they are not an eigenbasis. With $y_1 = \sqrt{3}\,|0\rangle/2$, $y_2 = |1\rangle/2$, the matrix

$$u = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

gives $x_1 = \eta_+/\sqrt{2}$, $x_2 = \eta_-/\sqrt{2}$. Their two outer products cancel the off-diagonal entries and recover ρ. Its unequal eigenvalues do not remove the ensemble freedom.

We can also *construct* this u using the converse proof. In the physical eigenbasis $|0\rangle, |1\rangle$, with eigenvalues $3/4, 1/4$, the spectral list y gives $w_1 = (1, 0)$, $w_2 = (0, 1)$. The other list gives $v_1 = (1, 1)/\sqrt{2}$, $v_2 = (1, -1)/\sqrt{2}$. These are already complete ONBs of the two-dimensional label space. Thus $u = \sum_{a=1}^{2} |v_a\rangle\langle w_a|$ has columns v_1, v_2, exactly the displayed matrix. On the physical system no unitary can turn the orthogonal pair $|0\rangle, |1\rangle$ into η_+, η_-, whose overlap is $1/2$.

Padding is substantive. The one-entry list (ψ) and the two-entry list $(\psi/\sqrt{2}, \psi/\sqrt{2})$ generate the same pure state. Pad the first to $(\psi, 0)$; the same 2×2 matrix gives the second list. There cannot be a rectangular isometry in both directions between label spaces of unequal dimensions. The zero padding makes the theorem's square unitary statement valid. The theorem above does not assert that the same completion argument works for two countably infinite lists.

Self-check 2.4

1. Use the proof's recipe to construct a unitary relating $y = (\psi, 0)$ to $x = (\psi/\sqrt{2}, \mathrm{i}\psi/\sqrt{2})$, with $\|\psi\| = 1$.
2. Why does a two-entry ensemble of one pure state not contradict extremality?

3. In the proof, where is finiteness of the ensembles used?

2.5 A laser mode without a phase reference

Model a single optical mode as the oscillator of *QM II*, Lecture 8. Its Hermite basis vectors h_n are now called number states: a number reading gives n photons. The coherent states from its Theorem 8.10 are

$$\psi_\alpha = e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} h_n .$$

The phase of α is not a global phase multiplying the whole vector. Replacing α by $\alpha e^{i\phi}$ multiplies the coefficient of h_n by $e^{in\phi}$: the relative phases of different number components change. Multiplying the whole vector by one common phase would leave its pure state unchanged and could not produce a mixture. Fix $\alpha \in \mathbb{C}$ and average its phase uniformly. A precise meaning of this average is that every bounded A has expectation $\frac{1}{2\pi}\int_0^{2\pi} \langle \psi_{\alpha e^{i\phi}}, A\psi_{\alpha e^{i\phi}} \rangle \, d\phi$. This is an ordinary integral of a continuous scalar function; we will justify the limiting series directly.

Proposition 2.6 (The phase-averaged state)**.** *In the sense of equality of all bounded expectations,*

$$\frac{1}{2\pi}\int_0^{2\pi} |\psi_{\alpha e^{i\phi}}\rangle\langle\psi_{\alpha e^{i\phi}}| \, d\phi = \rho_{\mathrm{av}} := e^{-\bar{n}} \sum_{n=0}^{\infty} \frac{\bar{n}^n}{n!} |h_n\rangle\langle h_n| , \qquad \bar{n} = |\alpha|^2 . \tag{2.3}$$

Proof. In a matrix element between h_m and h_n, the integrand is

$$e^{-|\alpha|^2} \frac{\alpha^m \bar{\alpha}^n}{\sqrt{m!n!}} e^{i(m-n)\phi} .$$

Its phase average vanishes for $m \neq n$ and is $e^{-\bar{n}}\bar{n}^n/n!$ for $m = n$. These diagonal entries are nonnegative and sum to one by the exponential series, so their sum defines a density operator by L1. For $\alpha = 0$ it is simply $|h_0\rangle\langle h_0|$.

To justify the average for every bounded A, truncate each coherent vector at level N, calling the result $s_N(\phi)$. Its squared error is

$$t_N = e^{-\bar{n}} \sum_{n>N} \frac{\bar{n}^n}{n!} \longrightarrow 0,$$

independently of ϕ, and $\|s_N(\phi)\| \leq 1$. Write $\psi = \psi_{\alpha e^{i\phi}}$. Adding and subtracting $\langle s_N, A\psi\rangle$ gives

$$\langle \psi, A\psi\rangle - \langle s_N, As_N\rangle = \langle \psi - s_N, A\psi\rangle + \langle s_N, A(\psi - s_N)\rangle .$$

Cauchy–Schwarz therefore bounds the absolute value by

$$|\langle \psi, A\psi\rangle - \langle s_N, As_N\rangle| \leq \|A\| \, (\|\psi\| + \|s_N\|) \, \|\psi - s_N\| \leq 2 \, \|A\| \, \sqrt{t_N}.$$

Thus the expectations converge uniformly in phase from continuous finite sums; the scalar integral exists and its error obeys the same bound. For a finite sum the matrix-element computation above gives exactly $e^{-\bar{n}} \sum_{n\leq N} \bar{n}^n \langle h_n, Ah_n\rangle / n!$. The missing diagonal tail has absolute value at most $\|A\| \, t_N$. Taking $N \to \infty$ proves the identity for every bounded A. ∎

Definition 2.4 covers discrete lists; the phase integral uses a continuous family of pure states. Its identity is not an application of finite HJW, but a separate calculation with explicit convergence control. For $\bar{n} > 0$, every Poisson weight is positive, so ρ_{av} has infinite rank and is mixed by Theorem 2.2. At $\bar{n} = 0$ it remains the pure vacuum state. Its number probabilities are Poisson probabilities $p_n = e^{-\bar{n}}\bar{n}^n/n!$. Shifting the index in the convergent series gives $\sum_n np_n = \bar{n}$ and $\sum_n n(n-1)p_n = \bar{n}^2$, hence the number-reading variance is $\bar{n}$. These are finite moments of the discrete reading distribution; the number operator is unbounded, so they are not automatic substitutions into the bounded-observable variance theorem.

For example, at $\bar{n} = 4$ the probabilities of vacuum and one photon are $e^{-4} \simeq 0.0183$ and $4e^{-4} \simeq 0.0733$. A mixture of number states with those Poisson weights gives exactly the same probabilities and all other bounded expectations as the uniform phase ensemble. This does not make it the thermal state: at the same mean number, L1's thermal distribution has vacuum probability $1/(1+\bar{n}) = 0.2$.

The physics

A photon-counting description of a laser mode often uses Poisson weights, while a calculation of interference uses a coherent amplitude and a phase. Equation (2.3) explains one precise equivalence between these descriptions. It does not derive the state produced by a particular laser. Mølmer's 1997 paper calls the absence of optical coherences in optics experiments a conjecture. His 1998 reply discusses ordinary discharge lasers and the entanglement of the pump electrons with the lasing atoms; it also explicitly allows optical coherence in principle. Those are physical arguments about sources. The mathematical equality here neither settles them nor forbids phase-sensitive measurements with a reference.

Uniform averaging is an additional assumption about how phase is represented when the reference is unavailable. A fixed-phase coherent state and ρ_{av} are generally different states. For example, when $\alpha > 0$ is real, the bounded operator $A = |h_0\rangle\langle h_1| + |h_1\rangle\langle h_0|$ has mean $2e^{-\alpha^2}\alpha$ in ψ_α, but zero in ρ_{av}. Thus the lesson is not that phase can never matter. It is that the *phase-averaged* state has two ensemble descriptions. A reference or a retained preparation record changes what information is available and must be included in a full model.

Self-check 2.5

1. Why do the off-diagonal Hermite matrix elements disappear in the uniform phase average?
2. At mean number one, compare the vacuum probabilities of the Poisson and thermal states.
3. Does finite HJW prove (2.3)? Identify the separate estimate that justifies its limit.

What does a state say about its preparation?

A normal state fixes all bounded expectations. It is pure exactly when it is a ray, and mixed states can have many ensembles, even with distinct eigenvalues. Finite ensembles of the same state are related by a unitary on their padded label lists. No state on a space of dimension at least two is sharp for every bounded observable. The phase-averaged coherent model and a Poisson number mixture are one state; this does not select a preparation history or determine how an actual laser works.

Summary. Pure means extreme; in dimension at least two it does not mean dispersion-free. A spectral decomposition is one preparation description, not a uniquely privileged one. HJW makes the finite-ensemble freedom exact; the laser identity identifies a uniform phase family and a discrete number-state ensemble with one state by a separate calculation. Later, the same care about what a density operator does not specify will be essential when we discuss reduced states and decoherence. Next we ask whether probabilities assigned directly to events or effects already force the trace rule.

Exercises. Try Sheet 2 in the separate *Exercise sheets* booklet (English).

Self-check answers.

2.1 1. No: the two states are identical. 2. Positivity need not survive a negative coefficient; for orthogonal rank-one states $2\,|0\rangle\langle 0| - |1\rangle\langle 1|$ has a negative expectation in $|1\rangle$.

2.2 1. The squared norm of $\sqrt{\rho_i}v$ is zero; applying the square root again gives $\rho_i v = 0$. 2. No: more than one positive eigenvalue gives the displayed nontrivial decomposition.

2.3 1. Project onto $(\psi + \varphi)/\sqrt{2}$ with unit $\varphi \perp \psi$. 2. No; the event depends on the state.

2.4 1. The first label vectors are $w_1 = (1, 0)$ and $v_1 = (1, \mathrm{i})/\sqrt{2}$. Complete with $w_2 = (0, 1)$, $v_2 = (\mathrm{i}, 1)/\sqrt{2}$. One valid unitary is $u = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & \mathrm{i} \\ \mathrm{i} & 1 \end{pmatrix}$. 2. Both positive-weight entries must describe the same ray. 3. The common range is finite-dimensional and the two label frames are completed to ONBs of the same finite $\mathbb{C}^m$.

2.5 1. The integral of $e^{\mathrm{i}(m-n)\phi}$ is zero for distinct integers m, n. 2. e^{-1} and $1/2$. 3. No; the phase-uniform error bound $2\,\|A\|\,\sqrt{t_N}$ controls the finite-vector approximation, and $\|A\|\,t_N$ controls the diagonal tail.

Sources. Busch–Lahti–Pellonpää–Ylinen, *Quantum Measurement* (2016), Thm 9.1, pp. 193–194; Example 9.1, p. 195; Thm 9.2, p. 196 (a more general decomposition statement, with its own kernel hypothesis). Nielsen–Chuang, *Quantum Computation and Quantum Information* (2010), Thm 2.6, pp. 103–104 (finite unitary freedom and zero padding). Landsman, *Foundations of Quantum Theory* (2017), pp. 46–48 (finite-dimensional version). K. Mølmer, *Phys. Rev. A* **55**, 3195–3203 (1997), especially pp. 3195–3197; reply, **58**, 4247 (1998). The coherent-state expansion is QM-II, Theorem 8.10.